

Extra study and consolidation in Algebra

Thobias Høivik

Contents

1	What is this?	3
2	Repeating some stuff about ideals, exact sequences and so forth	3

1 What is this?

Due to my being very interested in algebra, and interested in becoming very good at it, I have these extra notes for me to work through a lot of problems. Basically, there will probably not be any pedagogical value here.

2 Repeating some stuff about ideals, exact sequences and so forth

Problem 2.1. *Let*

$$R = \frac{k[x, y]}{(xy)}$$

and write $\bar{x}, \bar{y}$ for the residue classes.

What is $(\bar{x})$ as an ideal of R ?

Compute

$$R/(\bar{x}), \quad R/(\bar{y}), \quad R/(\bar{x}, \bar{y}).$$

Then classify them as prime/maximal/neither.

Sol. We note that the ideal generated by $\bar{x} = x + (xy)$ is

$$(\bar{x}) = \{\overline{xf} \mid f \in k[x, y]\}.$$

We have $(xy) \subseteq (x)$ and $(xy) \subseteq (y)$ as ideals of $k[x, y]$.

Consider the projection $\pi : k[x, y] \rightarrow R$ by

$$\pi(a) = \bar{a} = a + (xy).$$

Since π is surjective and $\{x\} \subseteq k[x, y]$ we have

$$\pi[(x)] = (\pi x) = (\bar{x}),$$

(note: surjectivity guarantees that scalar multiplication in R can be pulled back to $k[x, y]$, ensuring $\pi[(x)] \supseteq (\pi(x))$) but also

$$\pi[(x)] = \frac{(x)}{(xy)}$$

since $(xy) \subseteq (x)$.

Similarly,

$$\frac{(y)}{(xy)} = \pi[(y)] = (\bar{y}),$$

and

$$\frac{(x, y)}{(xy)} = \pi[(x, y)] = (\bar{x}, \bar{y}).$$

Thus we can apply the third isomorphism theorem to get

$$\frac{R}{(\bar{x})} = \frac{k[x, y]/(xy)}{(x)/(xy)} \cong \frac{k[x, y]}{(x)} \cong k[y],$$

and

$$\frac{R}{(\bar{y})} = \frac{k[x, y]/(xy)}{(y)/(xy)} \cong \frac{k[x, y]}{(y)} \cong k[x],$$

and

$$\frac{R}{(\bar{x}, \bar{y})} = \frac{k[x, y]/(xy)}{(x, y)/(xy)} \cong \frac{k[x, y]}{(x, y)} \cong k.$$

The first two are integral domains so $(\bar{x}), (\bar{y})$ are prime, though not maximal, while $(\bar{x}, \bar{y})$ is maximal since $R/(\bar{x}, \bar{y}) \cong k$ is a field. $\square$

Problem 2.2. Let

$$R = \frac{k[x, y]}{(xy)},$$

and let

$$\pi : k[x, y] \rightarrow R$$

be the quotient map.

Prove that every prime ideal $P \subseteq R$ contains $\bar{x}$ or $\bar{y}$.

Then, classify every prime ideal of R .

You may use without proof:

$$P \subseteq R \text{ prime} \Leftrightarrow \pi^{-1}(P) \subseteq k[x, y] \text{ is prime containing } (xy).$$

Proof. We have $\overline{xy} = 0$, and since P is an ideal, and thus a group, we have $0 \in P$, and since P is prime we must then have

$$\bar{x} \in P \text{ or } \bar{y} \in P.$$

Suppose P contains $\bar{x}$. Then, $(\bar{x}) \subseteq P$.

Then, by the third isomorphism theorem,

$$\frac{R/(\bar{x})}{P/(\bar{x})} \cong R/P,$$

but

$$\frac{R}{(\bar{x})} \cong k[y]$$

so

$$R/P \cong \frac{k[y]}{P/(\bar{x})}.$$

Since P is a prime ideal of R , the quotient R/P is an integral domain. Therefore, $P/(\bar{x})$ must be a prime ideal of $k[y]$.

Since $k[y]$ is a Principal Ideal Domain (PID), its prime ideals are:

1. The zero ideal (0) , or
2. Irreducible polynomial ideals $(f(y))$, where $f(y) \in k[y]$ is an irreducible polynomial.

Under the canonical quotient map $R \rightarrow R/(\bar{x}) \cong k[y]$, these correspond precisely to the prime ideals of R containing $\bar{x}$:

1. $P = (\bar{x})$, corresponding to $(0) \subset k[y]$.
2. $P = (\bar{x}, f(\bar{y}))$, where $f(y) \in k[y]$ is irreducible.

By symmetric reasoning, if $\bar{y} \in P$, then $P/(\bar{y})$ is a prime ideal of $k[x]$, which yields:

1. $P = (\bar{y})$, corresponding to $(0) \subset k[x]$.
2. $P = (\bar{y}, g(\bar{x}))$, where $g(x) \in k[x]$ is irreducible.

Combining both cases, the prime ideals of R consist of:

- Minimal prime ideals: $(\bar{x})$ and $(\bar{y})$.
- Maximal ideals: $(\bar{x}, f(\bar{y}))$ for irreducible $f(y) \in k[y]$, and $(\bar{y}, g(\bar{x}))$ for irreducible $g(x) \in k[x]$.

□

Problem 2.3. Let R be a commutative ring, I an ideal of R , and M an R -module.

You have three objects

$$IM, M, M/IM.$$

Part A.

Identify the two most natural R -linear maps between these objects and use them to construct a short exact sequence.

Give the maps explicitly and check exactness at every nonzero term.

Part B.

Now consider R/I as an R -module.

There is a canonical isomorphism

$$M \otimes_R R/I \cong M/IM.$$

Answer:

- What should the natural map

$$M \times R/I \rightarrow M/IM$$

probably be?

- Why is there an immediate well-definedness issue?

- What property would allow that bilinear map to induce a map from

$$M \otimes_R R/I?$$

Part C.

Suppose $N \subseteq M$. There is an obvious short exact sequence

$$0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0.$$

Tensor all with R/I .

You might hope to get

$$0 \rightarrow N \otimes_R R/I \rightarrow M \otimes_R R/I \rightarrow (M/N) \otimes_R R/I \rightarrow 0.$$

Is that always exact?

Proof of A. IM is the collection of finite sums of $i_j m_j$ where $i_j \in I$ and $m_j \in M$.

M/IM is then the collection of

$$\bar{x} = x + IM, \quad x \in M,$$

meaning the residue classes of elements in M modulo IM .

The most natural map from M to M/IM is the projection $\pi : M \rightarrow M/IM$,

$$\pi(x) = \bar{x}$$

sending elements x into their residue classes.

The natural map from IM to M is the inclusion $\iota : IM \rightarrow M$ simply sending

$$\iota(x) = x$$

since $IM \subseteq M$.

R -linearity of ι is essentially trivial since

$$\iota(rx + y) = rx + y = r\iota(x) + \iota(y).$$

For π observe that

$$\begin{aligned} \pi(rx + y) &= \overline{rx + y} \\ &= (rx + y) + IM \\ &= (rx + IM) + y + IM \\ &= r(x + IM) + (y + IM) \\ &= r\bar{x} + \bar{y} \\ &= r\pi(x) + \pi(y). \end{aligned}$$

This gives a natural sequence

$$0 \rightarrow IM \xrightarrow{\iota} M \xrightarrow{\pi} M/IM \rightarrow 0.$$

Then

$$\ker \pi = IM$$

since elements in IM are 0 modulo IM when you take the quotient, and $IM \subseteq M$.

Then

$$\text{im } \iota = IM,$$

since ι maps x to x with $IM \subseteq M$.

We have $\text{im } \pi = M/IM$ since a residue class $\bar{x} \in M/IM$ has at least one representative $x \in M$.

The only map from $M/IM \rightarrow 0$ is the trivial map, so

$$\ker(M/IM \rightarrow 0) = M/IM.$$

For the other side of the sequence there is once again only the trivial map from $0 \rightarrow IM$ with $\text{im}(0 \rightarrow IM) = 0$, and

$$\ker \iota = 0,$$

since $\iota(x) = 0_M$ implies $x = 0_M$, but as $IM \subseteq M$ we have that $0_{IM} = 0_M$.

Thus we have shown exactness at each point in the sequence, so this short sequence is exact. $\square$

Proof of B. A natural map $\beta : M \times R/I \rightarrow M/IM$ might be

$$(m, \bar{x}) \mapsto \overline{mx},$$

meaning

$$(m, r + I) \mapsto (mr) + IM$$

Notice that $(m, \bar{x}) \mapsto \overline{m+x}$ would very obviously create issues if we try to apply it to the tensor product since we require that $m \otimes 0$ is 0. Furthermore, $m+r$ is generally not an element of M for arbitrary $r \in R$.

For well-definedness suppose $r + I = s + I$. Then $r - s \in I$.

Then $(r - s)m \in IM$ since $r - s \in I$ and $m \in M$.

Thus we have that

$$\begin{aligned} \beta(m, r) - \beta(m, s) &= (mr + IM) - (ms + IM) \\ &= (m(r - s)) + IM, \end{aligned}$$

but $m(r - s) \in IM$ so this is 0_{IM} .

Lastly

$$\begin{aligned} \beta(m_1 + m_2, \bar{r}) &= ((m_1 + m_2)r) + IM \\ &= (m_1r + m_2r) + IM \\ &= (m_1r + IM) + (m_2r + IM) \\ &= \beta(m_1, \bar{r}) + \beta(m_2, \bar{r}), \end{aligned}$$

and

$$\begin{aligned}
\beta(m, \bar{r} + \bar{s}) &= (m(r + s)) + IM \\
&= (mr + ms) + IM \\
&= (mr + IM) + (ms + IM) \\
&= \beta(m, \bar{r}) + \beta(m, \bar{s}),
\end{aligned}$$

and

$$\begin{aligned}
\beta(am, \bar{r}) &= ((am)r) + IM \\
&= (m(ar)) + IM \\
&= \beta(m, a\bar{r}).
\end{aligned}$$

Thus, by the universal property of the tensor product, we can factor through a unique map so that

$$\begin{array}{ccc}
M \times R/I & \xrightarrow{\beta'} & M/IM \\
& \searrow & \uparrow \exists! \\
& & M \otimes_R R/I
\end{array}$$

commutes. □

Proof of C. We begin with the short exact sequence

$$0 \rightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \rightarrow 0.$$

Tensoring with R/I induces maps $\iota_R : N \otimes_R R/I \rightarrow M \otimes_R R/I$ sending

$$n \otimes \bar{r} \mapsto n \otimes \bar{r}$$

by

$$\iota_R(n \otimes \bar{r}) = \iota(n) \otimes \bar{r},$$

and $\pi_R : M \otimes_R R/I \rightarrow (M/N) \otimes_R R/I$ sending

$$m \otimes \bar{r} \mapsto \overline{m} \otimes \bar{r}$$

by

$$\pi_R(m \otimes \bar{r}) = \pi(m) \otimes \bar{r} = (m + N) \otimes \bar{r}.$$

In general tensoring is right exact, meaning

$$M' \otimes L \rightarrow M'' \otimes L \rightarrow M''' \otimes L \rightarrow 0$$

would remain exact if

$$M' \rightarrow M'' \rightarrow M''' \rightarrow 0$$

was.

Take for instance

$$0 \rightarrow 2\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

and tensor with $\mathbb{Z}/2\mathbb{Z}$.

Then we imagine we get an exact sequence

$$0 \rightarrow 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

However, there is no such exact sequence, with the

$$0 \rightarrow 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$$

term being the problem.

Consider $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$. Take $2 \otimes \bar{1}$. Then the image is $2 \otimes \bar{1}$, but in $\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}$ we have

$$(2 \cdot 1) \otimes \bar{1} = 1 \otimes (2\bar{1}) = 0.$$

In $2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}$, however, $2 \otimes \bar{1}$ is nonzero. Take the $\mathbb{Z}$ -bilinear map $B: 2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}_2$ given by $B(2m, \bar{n}) = mn \pmod{2}$ and take the unique map factoring through the tensor product $\beta: 2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}_2$.

As an asside, the map B is well defined since taking the product of two integers modulo 2 gives an element in $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$.

It is linear since

$$\begin{aligned} B(2m + 2m', n) &= (2m + 2m')n \pmod{2} \\ &= 2mn + 2m'n \pmod{2} \\ &= B(2m, n) + B(2m', n) \end{aligned}$$

in $\mathbb{Z}/2\mathbb{Z}$. The argument for linearity in the second coordinate and for scalar multipliacion from $\mathbb{Z}$ on the coordinates is analogous.

Then

$$\beta(2 \otimes 1) = 1 \neq 0.$$

Thus there is a nonzero element sent to 0, so

$$\ker(\iota_R) \neq 0.$$

□